

Sample from Personal Research Project

This sample demonstrates my analysis of heterogeneous treatment effects, presented as a research update to my faculty supervisor. It is from my personal research project, assisted by a UC Davis professor, using that professor's data from his own projects. My project asks what factors influence post-treatment willingness to pay for two different types of air quality forecasting sources in Punjab, Pakistan (EPD forecasting and PAQI forecasting). In the professor's experiment, he assigned households either the EPD forecasting or the PAQI forecasting. One of my findings was that treatment itself is the biggest determinant of relative willingness to pay.

In this sample, I investigate heterogeneous treatment effects using causal forests. This sample is part of an email update I sent to my professor.

Sample begins on following page.

Definitions and Purpose

I analyzed heterogeneous treatment effects for air quality forecasting using causal forests.

EPD = Environmental Protection Department, a government-run department.

PAQI = Pakistan Air Quality Initiative, a citizen run program.

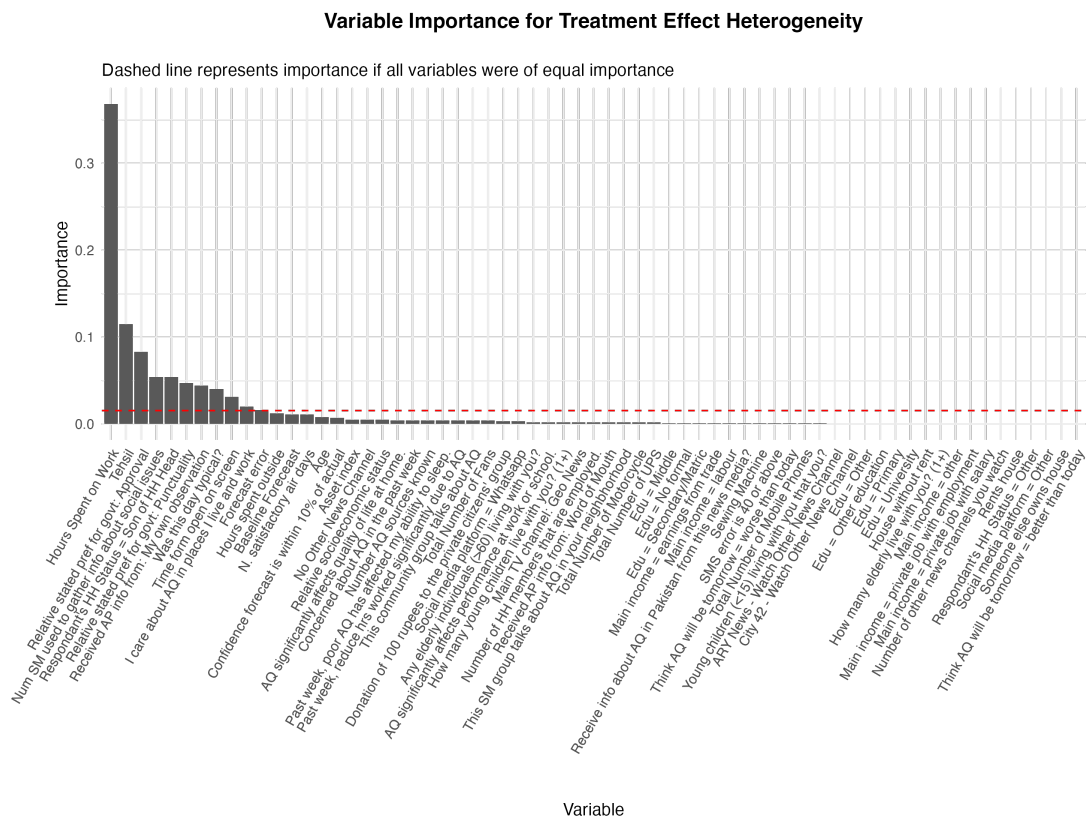
Treatment = attribution to EPD (vs. PAQI).

Outcome = relative WTP, coded as $WTP_PAQI - WTP_EPD$.

All treatment effects are negative, meaning EPD attribution consistently reduces relative WTP (increases preference for EPD/government forecasting). The analysis just tries to identify which household characteristics make this effect stronger.

Variable importance

Below, I plot all variables in order of their importance assigned by causal forests.



Three variables account for the majority of treatment effect heterogeneity: hours spent on work (importance = 0.40), location/tehsil (importance = 0.10), and baseline relative government approval (importance = 0.05). Together, these explain 55% of variation in treatment effects.

Top Three Variables

Here, I will briefly look more in-depth at the top three variables with bar plots.

The orange dashed line in the bar plots is the (negative) average of willingness to pay over PAQI and EPD forecasting.

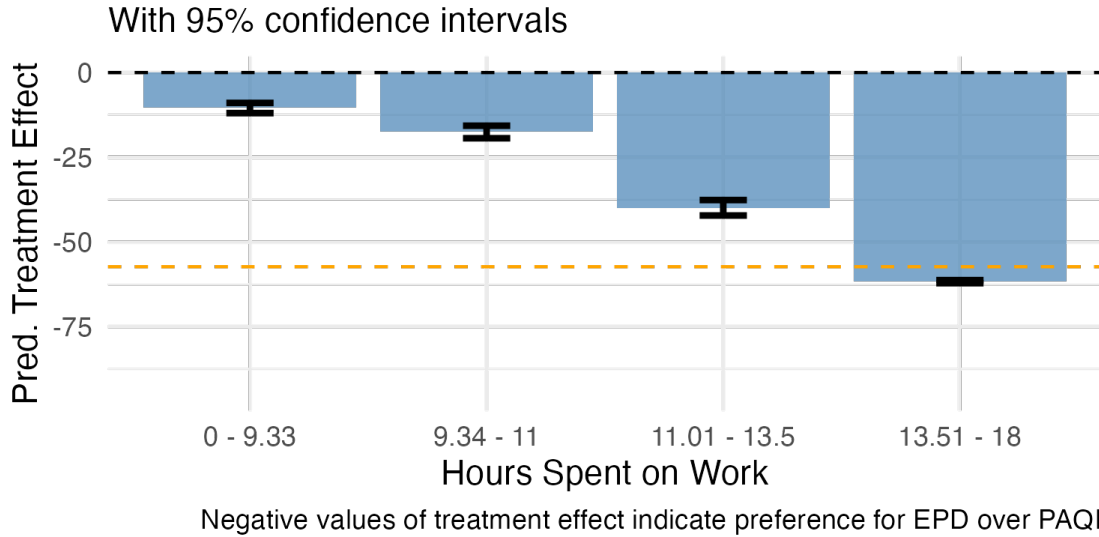
Work Hours

I first run a best linear projection on work hours as it is the most important variable, according on causal forests.

Table 1: Best linear projection of work hours on treatment

Term	Estimate	Std. Error	Statistic	P-Value
(Intercept)	56.976	5.399	10.552	0
Work Hours	-7.778	0.454	-17.136	0

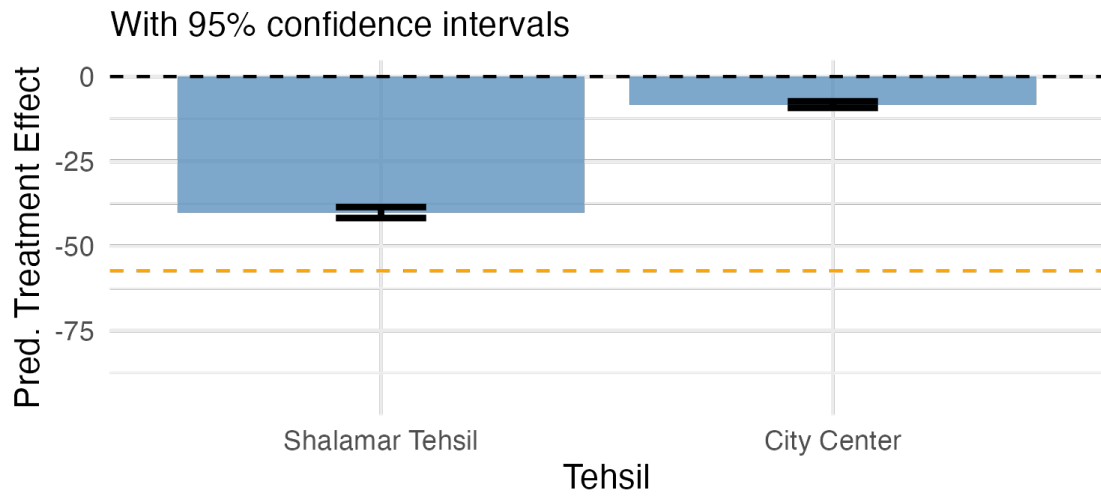
Each additional hour worked per week decreases the treatment effect by 8.3 PKR ($p < 0.001$). This means households working more hours develop substantially stronger preferences for EPD attribution.



This plot also reflects that as work hours increase, treatment effect becomes more negative (more preference for EPD forecasting). Work hours is potentially a proxy for income, as the survey did not include a direct question about income. The closest question was asset index, which is still fundamentally different than income and may actually be a worse proxy than work hours.

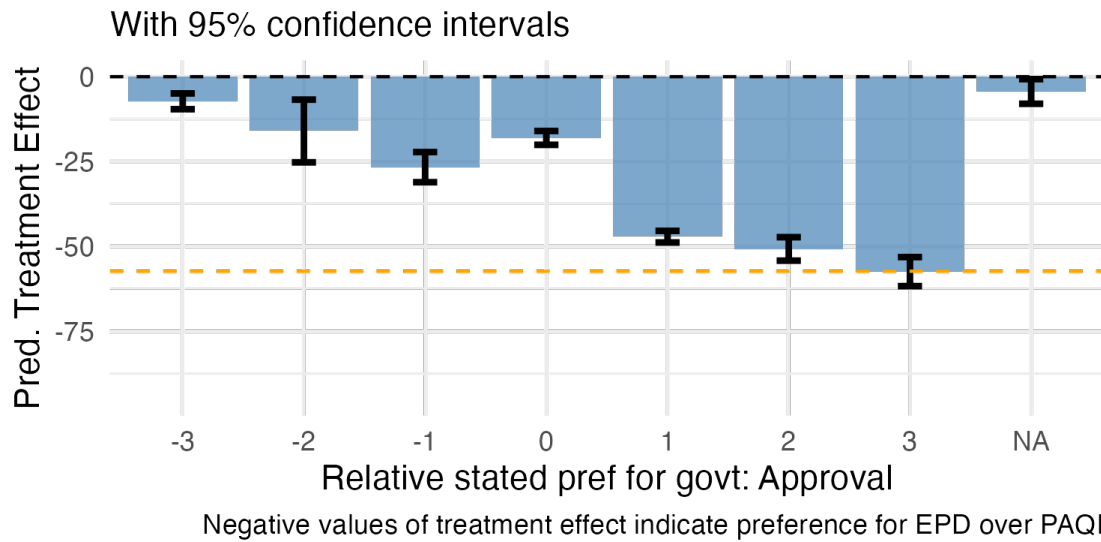
This would make sense as the professor's own experience in Pakistan is that lower-income individuals have a much lower trust in the government (recall EPD is a government department).

Location



People living in Shalamar have a much more negative treatment effect. If people living in Shalamar have lower incomes then this aligns with the theory about work hours being a proxy for income.

Approval Index



People with a positive approval for the government are more receptive to the EPD treatment. This is fairly intuitive.